

INVESTIGATION REPORT

Issue 196: Interrupt during single-step retired a handler instruction

openhwgroup/cv32e40x

Document	RTL-TRIAGE-CV32E40X-196
Disposition	Bug confirmed, fix verified
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40x#196
Issue type	Bug report from an integrator or user
Affects	cv32e40x
Configuration	pre-fix RTL adcb717b (2021-08-20) vs master d952cd6; X_EXT=0
Disposition	Bug confirmed, fix verified
Severity	Not assessed
Agreement	Same disposition as the independent human record.
Evidence	6 cited, 5 present, 1 not retained

Disposition meaning. The reported behaviour was reproduced, and the checks listed under Evidence pass against the corrected design.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

Blind hypothesis from the report and the debug-spec quote: while single-stepping (dcsr.step=1, not in debug mode), once an interrupt is taken the core must re-enter debug mode before any instruction of the handler retires. Property (bound to cv32e40x_controller_fsm): a flag is armed when ctrl_fsm_o.irq_ack fires with step=1 outside debug mode and cleared in DEBUG_TAKEN; while armed, wb_valid_i must be 0. Pre-fix mechanism read from the RTL: pending_single_step = !debug_mode_q && step && wb_valid_i, and the FUNCTIONAL state evaluates the interrupt branch before the single-step check, so when the stepped instruction retires and an interrupt is pending in the same cycle the interrupt is taken, the pipeline is killed, and pending_single_step is only raised again when the handler's first instruction retires. The pre-fix RTL fails the property at BMC step 22 (106 s): dret at step 17, interrupt acknowledged at step 17 with step=1, handler instruction retires at step 21 (wb_valid_i=1, armed=1), DEBUG_TAKEN only at step 22. On master the property passes unbounded (abc pdr 12.6 s) and BMC-30; the covers 'interrupt taken while stepping' (step 13) and 'debug entry after that' (step 13) are reached, so the proof is not vacuous. Environment: quiet bus (no instr/data errors, so no NMI path). Human fix: PR #271 (bf832e77, 2021-10-12) changes pending_single_step to (!debug_mode_q && dcsr_i.step && (wb_valid_i || ctrl_fsm_o.irq_ack)) && !pending_debug; the irq_ack term is precisely the event the checker arms on, so debug is entered in the cycle after the interrupt with dpc = handler address. The report's LSU corner case (interrupt not allowed while an LSU instruction is in WB) is outside the property: there the interrupt is deferred to the next step, which is spec-compliant.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Checker used

```
checker_196_common.sv (+ checker_196_prefix.sv / checker_196_master.sv bind lines: debug_single_step_i vs dcsr_i.step)
```

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	counterexample	formal/prefix_bmc/job/engine_0/trace.vcd	not kept	pre-fix: a_196_no_retire_before_debug fails at step 22: irq_ack with step=1 at 17, wb_valid_i=1 at 21 while armed, DEBUG_TAKEN at 22 (cex_prefix_handler_retires_in_step.png)
E2	proof	formal/prefix_bmc/sby.log	present	smtbmc yices BMC depth 30 FAIL (106 s) on adcb717b
E3	proof	formal/prefix_cover/sby.log	present	pre-fix cover: step-retire at 15, irq-in-step at 18, debug-after-irq at 20 (all reached)
E4	proof	formal/master_prove/sby.log	present	master: abc pdr PASS unbounded (12.6 s)
E5	proof	formal/master_bmc/sby.log	present	master: BMC depth 30 PASS (89.9 s)
E6	proof	formal/master_cover/sby.log	present	master cover: irq-in-step and debug-after-irq reached at step 13, step-retire at 14 -> not vacuous

Paths are relative to runs/e40x-196. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.
- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.

Artefact(s) E1 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- The fix is verified against the checks listed under Evidence only. This is not a regression sign-off and does not cover the wider test suite.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory

runs/e40x-196

Title source

Title and configuration read back from reports/issues/cv32e40x_issue_196.pdf, the previously published report for this issue.

Inputs shown	title, body(links stripped), labels
Human record	PR #271 (bf832e77, 2021-10-12): pending_single_step also asserted on ctrl_fsm_o.irq_ack so an interrupt taken during a step enters debug immediately with dpc = handler
Agent confidence	0.9. This is the investigating agent's own estimate. It is not a measurement and not a reliability figure.
Iterations	1
Run by	Claude directly (no headless agent)

Measured tool and step time

Step	Measured
read issue + pre-fix checkout + wrapper	1.5 min (90 s)
locate step/irq/debug logic in both revisions, hypothesis	1.8 min (110 s)
checker + two bind maps + pre-flight check	60.0 s
pre-fix BMC: CEX step 22	1.8 min (106 s)
master pdr + BMC + covers (parallel)	1.5 min (90 s)
read fix PR #271, compare	40.0 s
report	5.0 min (300 s)
Total	13.3 min (796 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

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